

# Higher-Form Symmetry in a Nutshell

From differential forms to the generalized Landau paradigm

August 31, 2026

*One-line nutshell:* a symmetry is a **topological operator**;  
the only thing we generalize is the **dimension of its support**.

# Plan

- 1 Motivation
- 2 Mathematical foundation
- 3 Ordinary symmetry, recast
- 4 Higher-form symmetry: definition
- 5 Examples
- 6 Spontaneous breaking and phenomena
- 7 Outlook

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# The Landau paradigm and its cracks

**Landau:** phases  $\leftrightarrow$  symmetries and their spontaneous breaking; critical modes = order-parameter fluctuations.

## Order without a local order parameter:

- **topological order** (toric code);
- **confinement / deconfinement**;
- the exactly massless **photon**.

## Claim of this talk

All three *are* symmetry-breaking phenomena — for symmetries acting on *loops and surfaces* instead of points.

# Roadmap

- ① **Language:** forms, homology, cohomology, linking.
- ② **Recast:** ordinary symmetry = topological codimension-1 operator.
- ③ **Generalize:**  $p$ -form symmetry = codimension- $(p+1)$  operators acting on  $p$ -dimensional objects.
- ④ **Examples:** Maxwell, toric code  $\rightarrow \mathbb{Z}_N$  gauge theory, Chern–Simons, Yang–Mills center.
- ⑤ **Phenomena:** generalized SSB, Goldstone photon, Mermin–Wagner, topological order, anomalies.

## Conventions

Spacetime dimension  $D = d + 1$ ;  $\Sigma_k, C_k$  closed oriented  $k$ -manifolds;  $d$  exterior derivative,  $\star$  Hodge star.

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# Differential forms and the wedge product

## Definition: $p$ -form

$$\alpha = \frac{1}{p!} \alpha_{\mu_1 \dots \mu_p} dx^{\mu_1} \wedge \dots \wedge dx^{\mu_p}, \quad \alpha_{\mu_1 \dots \mu_p} \text{ totally antisymmetric.}$$

**Wedge product**  $\wedge : \Omega^p \times \Omega^q \rightarrow \Omega^{p+q}$ ; in components,

$$(\alpha \wedge \beta)_{\rho_1 \dots \rho_{p+q}} = \frac{(p+q)!}{p! q!} \alpha_{[\rho_1 \dots \rho_p} \beta_{\rho_{p+1} \dots \rho_{p+q}]}$$

- Familiar example: the field strength  $F = \frac{1}{2} F_{\mu\nu} dx^\mu \wedge dx^\nu$ .
- A  $p$ -form is the natural object to **integrate over a  $p$ -dimensional surface**.

# The exterior derivative

$d : \Omega^p \rightarrow \Omega^{p+1}$ ; in components,

$$(d\alpha)_{\mu_0\mu_1\cdots\mu_p} = (p+1) \partial_{[\mu_0} \alpha_{\mu_1\cdots\mu_p]},$$

with the two structural properties

$$\boxed{d^2 = 0}, \quad d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^p \alpha \wedge d\beta.$$

- $p = 0$ : the gradient.     $p = 1$ :  $F = dA$  gives  $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ .
- Then  $dF = d^2A = 0$  automatically — the **Bianchi identity**. *Remember this: it will be a symmetry.*

# The Hodge star

A metric defines an inner product on  $p$ -forms and the map  $\star : \Omega^p \rightarrow \Omega^{D-p}$ :

$$\boxed{\alpha \wedge \star \beta = \langle \alpha, \beta \rangle \Omega,} \quad \langle \alpha, \beta \rangle = \frac{1}{p!} \alpha_{\mu_1 \dots \mu_p} \beta^{\mu_1 \dots \mu_p},$$

$$(\star \beta)_{\nu_1 \dots \nu_{D-p}} = \frac{1}{p!} \varepsilon_{\nu_1 \dots \nu_{D-p} \mu_1 \dots \mu_p} \beta^{\mu_1 \dots \mu_p} \quad (\Omega = d^D x \text{ the volume form}).$$

- In  $D = 4$  the dual of a 2-form is again a 2-form:  $F \leftrightarrow \star F$ .
- $\star$  turns a  $(p+1)$ -form current into a codimension- $(p+1)$  integrand — *this* is where codimension enters.

# Stokes' theorem

$$\int_V d\omega = \int_{\partial V} \omega$$

- $\partial^2 = 0$  (“a boundary has no boundary”) mirrors  $d^2 = 0$ .

## Slogan

$d$  and  $\partial$  are adjoint under integration. Everything “topological” in this talk is Stokes' theorem, used twice.

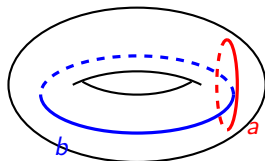
# Homology: which surfaces are “the same”?

$p$ -chains with boundary map  $\partial$ ,  $\partial^2 = 0$ :

- **cycles:**  $\partial C = 0$ ;    **boundaries:**  $C = \partial V$ .

$$H_p(M) = \ker \partial / \operatorname{im} \partial$$

- Deforming a cycle = adding a boundary:  
 $C' = C + \partial V$ .
- “Topological” will mean: **depends only on  $[C]$** .



$H_1(T^2) = \mathbb{Z}^2$ :  $a, b$  closed,  
bound nothing.

*This picture returns as the  
toric-code ground states.*

# de Rham cohomology and the pairing

Mirror construction with  $d$  in place of  $\partial$ :

$$H_{\text{dR}}^p(M) = \{\text{closed}\} / \{\text{exact}\}$$

**The pairing:** for  $d\omega = 0$ ,  $\partial C = 0$ ,

$$\langle [\omega], [C] \rangle = \int_C \omega \quad \text{well defined on classes:} \quad \int_{C+\partial V} \omega = \int_C \omega + \int_V d\omega = \int_C \omega.$$

## Take-home

“closed form over a cycle” = “number invariant under deforming the cycle”.  
Topological surface operators will be *exactly* this.

## Poincaré duality: a calculus of delta-forms

An oriented  $k$ -cycle  $Y_k \subset M_D$  has a distributional  $(D-k)$ -form  $\text{PD}(Y)$ , defined by

$$\int_Y \omega = \int_M \omega \wedge \text{PD}(Y) \quad \text{for all } \omega \in \Omega^k.$$

Integration by parts gives its one key identity:

$$\text{d PD}(Y_k) = (-1)^k \text{PD}(\partial Y_k) \quad \text{“d of a delta-form = delta-form of the boundary”}.$$

- $Y$  closed  $\Rightarrow \text{PD}(Y)$  closed;  $Y$  a boundary  $\Rightarrow \text{PD}(Y)$  exact: homology  $\rightarrow$  cohomology.
- This identity will *generate* symmetry defects and Ward identities in Part 4.

# Intersection and linking

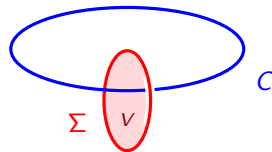
**Intersection number**  $\#(V, C) \in \mathbb{Z}$  of  $V_{D-p}$  and  $C_p$ :  
signed count of crossing points, computed by  $\int_V \text{PD}(C)$ .

**Linking number** of disjoint  $\Sigma_{D-p-1} = \partial V$  and  $C_p$ :

$$\text{Link}(\Sigma, C) = \#(V, C) = \int_V \text{PD}(C).$$

Purely topological; in  $D = 3$  it is the Gauss linking number.

*Linking is the geometry of **charge measurement**.*



$\text{Link}(\Sigma, C) = 1$ :  $\Sigma$  bounds the disk  $V$ , pierced once by  $C$ .

## Three take-homes from the math

- ① A  $p$ -form integrates over a  $p$ -dimensional surface.
- ② Stokes:  $\int_V d\omega = \int_{\partial V} \omega$ ;  $d^2 = 0 \leftrightarrow \partial^2 = 0$ .
- ③ “Topological” = depends only on the homology class; classes pair by intersection/linking.

*That is all the mathematics this talk needs.*

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# Noether, rewritten in forms

Continuous 0-form symmetry: current 1-form  $J$ ;  $\partial^\mu J_\mu = 0$  becomes

$$\boxed{d \star J = 0} \quad (\star J \text{ a closed } (D-1)\text{-form}).$$

Define, on any closed **codimension-1** surface,

$$Q(\Sigma) = \int_{\Sigma} \star J, \quad \boxed{U_{\alpha}(\Sigma) = e^{i\alpha Q(\Sigma)}.$$

On an equal-time slice this is the familiar  $e^{i\alpha Q}$  — but  $\Sigma$  can be *any* closed hypersurface in spacetime.

# Localizing the Noether variation

Promote the constant parameter to a compactly supported local one,  $\lambda(x)$ . Its variation of the action *defines* the current:

$$\delta_\lambda S = \int_M \lambda \wedge d\star J.$$

- Constant  $\lambda$ : the symmetry,  $\delta S = 0$  identically.
- General  $\lambda$ : a *Noether probe* — on shell, away from insertions,  $\delta_\lambda S = 0$  for all  $\lambda$  forces  $d\star J = 0$ .

A symmetry **defect** = concentrating  $d\lambda$  on a codimension-1 locus. Next slide: do it with a delta-form.

# The defect from a singular gauge parameter

Let the parameter jump across  $\Sigma = \partial V$ : take

$$\lambda_V = \alpha \text{PD}(V) \implies d\lambda_V = \pm \alpha \text{PD}(\Sigma) \quad (\text{the boundary identity, signs suppressed}).$$

Substituting into the localized variation:

$$\delta_{\lambda_V} S = \alpha \int_{\Sigma} \star J \implies \boxed{U_{\alpha}(\Sigma) = e^{i\delta_{\lambda_V} S} = \exp\left(i\alpha \int_{\Sigma} \star J\right)}.$$

One construction delivers, at once: current conservation, the topological symmetry defect, and (Part 4) its action on charged operators.

## Why the charge is topological

For homologous surfaces,  $\Sigma - \Sigma' = \partial V$ :

$$Q(\Sigma) - Q(\Sigma') = \int_{\partial V} \star J = \int_V d \star J = 0.$$

### Conservation = topological invariance

$U_\alpha(\Sigma)$  depends only on the deformation class of  $\Sigma$ , as long as it does not cross a charged insertion.

Sweeping  $\Sigma$  past a charged local operator picks up the phase:

$$U_\alpha(\Sigma) \mathcal{O}_q(x) U_\alpha(\Sigma)^{-1} = e^{i\alpha q} \mathcal{O}_q(x) \quad \text{when } \Sigma \text{ links } x.$$

# The conceptual pivot

## Modern definition of a symmetry

A global symmetry is a set of **topological operators**  $U_g(\Sigma)$  with fusion  $U_g U_h = U_{gh}$ , acting on charged objects by linking.

- Works for **discrete** symmetries too — no current needed.
- Ordinary symmetry: codimension-1 operators, charged **points**.

Now just change the numbers.

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# Definition of a $p$ -form symmetry

## Definition (GKSW 2014)

Topological operators  $U_g(\Sigma_{D-p-1})$ ,  $g \in G$ , on closed **codimension**-( $p+1$ ) surfaces, with  $U_g U_h = U_{gh}$ , acting on  $p$ -**dimensional** charged operators  $W(C_p)$  by linking.

| $p$ | symmetry operator on | charged operator  | typical charge         |
|-----|----------------------|-------------------|------------------------|
| 0   | codim-1 hypersurface | point (local op.) | particle               |
| 1   | codim-2 surface      | <b>line</b>       | Wilson / 't Hooft line |
| 2   | codim-3 submanifold  | surface           | membrane               |

## Continuous case: currents and the linking action

Conserved  $(p+1)$ -form current:

$$d \star J_{p+1} = 0, \quad U_\alpha(\Sigma_{D-p-1}) = \exp \left( i\alpha \int_\Sigma \star J_{p+1} \right)$$

— topological, by the same Stokes argument. Charge is measured by a **linking phase**:

$$U_\alpha(\Sigma) W_q(C) U_\alpha(\Sigma)^{-1} = e^{i\alpha q \text{ Link}(\Sigma, C)} W_q(C).$$

On a spatial slice,  $\Sigma$  and  $C$  *intersect*: the phase becomes  $e^{i\alpha q \#(\Sigma, C)}$ .

*The next four slides derive all of this from one identity: the action variation  $\delta_\lambda S$ .*

# The action variation $\delta S$ , for general $p$

Parameter of a  $p$ -form symmetry: a  **$p$ -form**  $\lambda_p$ . The global transformations are the *flat* ones,  $d\lambda = 0$  (compact  $U(1)$ : identified modulo forms with  $2\pi\mathbb{Z}$  periods).

## Local Noether identity (this *defines* $J$ )

For a compactly supported local parameter  $\lambda$ ,

$$\delta_\lambda S = \int_M \lambda \wedge d\star J = (-1)^{(p+1)(D-p)} \int_M \star J \wedge d\lambda.$$

- Flat  $\lambda$ :  $\delta_\lambda S = 0$  identically — the symmetry.
- Arbitrary  $\lambda$  (a *Noether probe*): on shell, away from charged insertions,  $\delta_\lambda S = 0$  forces  $d\star J = 0$ .
- Everything that follows is this identity evaluated on *singular* parameters.

## $\delta S$ on a delta-form parameter = the symmetry defect

Take  $\Sigma = \partial V$  and the singular parameter

$$\lambda_V = (-1)^{p(D-p)} \alpha \text{PD}(V) \quad \implies \quad d\lambda_V = (-1)^{p(D-p)+D-p} \alpha \text{PD}(\Sigma),$$

by the boundary identity  $d \text{PD}(V) = (-1)^{D-p} \text{PD}(\partial V)$ . Substituting into  $\delta_\lambda S$  — all signs cancel:

$$\delta_{\lambda_V} S = (-1)^{(p+1)(D-p)} \int_M \star J \wedge d\lambda_V = \alpha \int_M \star J \wedge \text{PD}(\Sigma) = \alpha \int_\Sigma \star J.$$

The path-integral insertion generated by this localized transformation:

$$U_\alpha(\Sigma) = e^{i \delta_{\lambda_V} S} = \exp \left( i \alpha \int_\Sigma \star J \right).$$

# Charged extended operators

## Definition: $p$ -form charge

$W_q(C_p)$  has charge  $q$  if a *flat* parameter acts by

$$\delta_\lambda W_q(C_p) = -iq \left( \int_{C_p} \lambda \right) W_q(C_p).$$

- Basic example: a  $p$ -form field with  $\delta_\lambda A_p = -\lambda$  gives  $W_q(C) = e^{iq \int_C A_p}$ .
- For compact  $U(1)$ ,  $q \in \mathbb{Z}$  in the standard normalization.
- In spacetime,  $W_q$  sweeps a  $(p+1)$ -dimensional worldvolume; charge conservation = its inability to end.

# The Ward identity, derived

Promote  $\lambda$  to an arbitrary test  $p$ -form; a change of variables in the path integral gives

$$0 = \langle \delta_\lambda W_q \rangle + i \langle \delta_\lambda S W_q \rangle \implies \int_M \lambda \wedge \langle d\star J W_q(C) \rangle = q \int_C \lambda \langle W_q(C) \rangle.$$

$\lambda$  arbitrary  $\Rightarrow$  the contact-term **Ward identity**

$$\langle d\star J W_q(C_p) \rangle = q \text{PD}(C_p) \langle W_q(C_p) \rangle.$$

Integrate over  $V$  ( $\partial V = \Sigma$ ): the delta-form gives  $\int_V \text{PD}(C) = \text{Link}(\Sigma, C)$ , hence

$$[Q(\Sigma), W_q(C)] = q \text{Link}(\Sigma, C) W_q(C) \xrightarrow{\text{exponentiate}} U_\alpha W_q U_\alpha^{-1} = e^{i\alpha q \text{Link}(\Sigma, C)} W_q.$$

## Structural fact: higher-form symmetries are abelian

### Proposition

For  $p \geq 1$ , an invertible  $p$ -form symmetry group is **abelian**.

*Argument.* Codimension  $\geq 2$  surfaces can be deformed around each other without intersecting, so the two orderings are topologically equivalent:

$$U_g(\Sigma_1) U_h(\Sigma_2) = U_h(\Sigma_2) U_g(\Sigma_1). \quad \square$$

- Contrast  $p = 0$ : codimension-1 walls order spacetime into before/after — non-abelian groups allowed.
- Non-abelian topological order will need *non-invertible* symmetry (outlook).

## Coupling to a background field

A continuous  $p$ -form symmetry couples to a **background**  $(p+1)$ -form gauge field:

$$S[\phi; B] = S[\phi; 0] + \int_M B_{p+1} \wedge \star J, \quad B \sim B + d\Lambda_p.$$

### Background gauge invariance $\Leftrightarrow$ conservation

$$\delta_\Lambda S = \int_M d\Lambda \wedge \star J = \pm \int_M \Lambda \wedge d\star J \quad \Rightarrow \quad \delta_\Lambda S = 0 \iff d\star J = 0.$$

- $Z[B]$  packages all current correlators. A failure of  $Z[B + d\Lambda] = Z[B]$  is an **'t Hooft anomaly** (Part 6), not a failure of conservation.
- Discrete symmetry:  $B$  is a *flat* cocycle,  $[B] \in H^{p+1}(M, G)$  — a closed defect network with no endpoints.

# Gauging a finite symmetry, and the dual symmetry

**Gauging**  $G^{(p)}$  ( $G$  finite abelian) = summing over all flat backgrounds, with a coupling to the dual group  $\widehat{G} = \text{Hom}(G, U(1))$ :

$$Z_{\text{gauged}}[\widehat{B}] = \sum_{[B] \in H^{p+1}(M, G)} \frac{Z[B]}{|\text{Aut}(B)|} \exp \left( 2\pi i \int_M \langle B \smile \widehat{B} \rangle \right).$$

On a closed  $M_D$  this pairing is *perfect*, with  $[\widehat{B}] \in H^{D-p-1}(M, \widehat{G})$ ; a  $q$ -form background has degree  $q+1 = D-p-1$ , so

$$G^{(p)} \xrightarrow{\text{gauging}} \widehat{G}^{(D-p-2)}.$$

- $D = 2$ : Ising  $\mathbb{Z}_2^{(0)} \rightarrow \mathbb{Z}_2^{(0)}$  — Kramers–Wannier.
- $D = 3$ : Ising  $\mathbb{Z}_2^{(0)} \rightarrow \mathbb{Z}_2^{(1)}$  — the toric code’s 1-form symmetry.

# One symmetry, three equivalent descriptions

|                                 |                                               |
|---------------------------------|-----------------------------------------------|
| topological-defect formulation: | $U_g(\Sigma_{D-p-1}), \quad U_g U_h = U_{gh}$ |
| current formulation:            | $d \star J_{p+1} = 0$                         |
| background-field formulation:   | $B_{p+1} \sim B_{p+1} + d\Lambda_p$           |

Common operator content:

$$U_g(\Sigma) W_\chi(C) U_g(\Sigma)^{-1} = \chi(g)^{\text{Link}(\Sigma, C)} W_\chi(C), \quad [H, U_g(\Sigma)] = 0$$

for a static spatial  $\Sigma$  — topological invariance in the time direction *is* conservation.

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## Example 1: free Maxwell theory in $D = 4$ (electric)

$$S = -\frac{1}{2e^2} \int F \wedge \star F, \quad F = dA.$$

**Electric 1-form**  $U(1)$  — the *equation of motion* is a conservation law:

$$d\left(\frac{1}{e^2} \star F\right) = 0 \quad \implies \quad J_e = \frac{1}{e^2} \star F \text{ is a closed 2-form.}$$

$$U_\alpha^e(\Sigma_2) = \exp\left(\frac{i\alpha}{e^2} \int_{\Sigma} \star F\right), \quad W_n(C) = e^{in \oint_C A},$$

$$U_\alpha^e(\Sigma) W_n(C) U_\alpha^e(\Sigma)^{-1} = e^{in\alpha \text{ Link}(\Sigma, C)} W_n(C).$$

$U_\alpha^e$  measures electric flux through  $\Sigma$ ; a Wilson line is a rigid unit electric flux line.



## Example 2: the toric code and its $e$ & $m$ symmetries

Qubits on the links of a square lattice ( $D = 2+1$ ):

$$H = - \sum_v A_v - \sum_p B_p, \quad A_v = \prod_{\ell \ni v} X_\ell, \quad B_p = \prod_{\ell \in \partial p} Z_\ell.$$

Two  $\mathbb{Z}_2$  1-form symmetries, supported on closed loops:

$$W^e(C) = \prod_{\ell \in C} Z_\ell \quad (\text{direct lattice}), \quad W^m(\hat{C}) = \prod_{\ell \perp \hat{C}} X_\ell \quad (\text{dual lattice}),$$

$$W^e(C) W^m(\hat{C}) = (-1)^{\#(C, \hat{C})} W^m(\hat{C}) W^e(C).$$

- Deformable by multiplying stabilizers  $\Rightarrow$  topological on the ground space.
- *Open* strings end on anyons:  $e$  (vertex defects),  $m$  (plaquette defects).
- Each symmetry's operator is the *other's* charged line — remember this for the anomaly discussion.

## Generalizing: $\mathbb{Z}_N$ gauge theory in $d + 1$ dimensions

Replace qubits by  $N$ -state qudits,  $ZX = \omega XZ$  with  $\omega = e^{2\pi i/N}$ , on a  $d$ -dimensional lattice. For a curve  $C$  and a codimension-1 surface  $M$  on the dual lattice:

$$V(C) = \prod_{\ell \in C} X_\ell, \quad U(M) = \prod_{\ell \perp M} Z_\ell, \quad \boxed{U(M) V(C) = \omega^{\#(C,M)} V(C) U(M).}$$

- $U(M)$  generates the electric  $\mathbb{Z}_N$  **1-form** symmetry; its charged lines are the flux strings  $V(C)$ .
- The magnetic partner is a  $(D-3)$ -form symmetry — both are 1-form precisely in  $D = 3$ : the toric code is that self-dual case.
- Continuum limit: BF theory  $S = \frac{N}{2\pi} \int b_{D-2} \wedge da_1$ , where  $[\int_M b, \oint_C a] = \frac{2\pi i}{N} \#(C, M)$  reproduces the same algebra.

# Ground-state degeneracy from the algebra

On the torus, wrap  $U$  and  $V$  on the two cycles with  $\#(C_x, C_y) = 1$  (the homology picture!).  
The algebra becomes **projective**:

$$U_x U_y = e^{2\pi i/N} U_y U_x.$$

- Every irrep of this algebra is  $N$ -dimensional  $\Rightarrow$   **$N$  ground states on  $T^2$** .
- Both operators are topological: the degeneracy is locally invisible — *topological order*.  
Anyons = endpoints of open charged lines.

## Example 3: $U(1)_k$ Chern–Simons theory

$$S = \frac{k}{4\pi} \int_{M_3} a \wedge da, \quad k \in \mathbb{Z}.$$

Equation of motion  $\frac{k}{2\pi} da = 0 \Rightarrow$  every Wilson line is topological:

$$W_n(C) = e^{in \oint_C a}, \quad W_n(C) W_m(C') = e^{2\pi i nm \text{Link}(C,C')/k} W_m(C') W_n(C).$$

- In  $D = 3$ , 1-form symmetry operators and charged objects are both *lines* — and here they coincide: the anyon lines generate a  $\mathbb{Z}_k$  1-form symmetry acting on themselves by braiding.
- On  $T^2$ : the same projective algebra as before  $\Rightarrow k$  ground states.
- Topological spin  $\theta_n = e^{i\pi n^2/k}$  is nontrivial  $\Rightarrow$  this  $\mathbb{Z}_k$  symmetry is *anomalous* (Part 6: statistics = anomaly).

## Example 4: $SU(N)$ center symmetry and confinement

Pure  $SU(N)$  Yang–Mills: gluons are adjoint  $\Rightarrow$  the center  $\mathbb{Z}_N \subset SU(N)$  acts trivially on every local operator. It survives as a discrete electric 1-form symmetry  $\mathbb{Z}_N^{(1)}$ , acting on the fundamental **Wilson loop**:

$$W(C) = \text{Tr}_F \mathcal{P} \exp \left( i \oint_C A \right), \quad U_k(\Sigma_2) W(C) U_k(\Sigma_2)^{-1} = e^{2\pi i k \text{Link}(\Sigma, C)/N} W(C).$$

### Confinement finally gets an order parameter

$$\begin{array}{lll} \langle W(C) \rangle \sim e^{-T \cdot \text{Area}(C)} & \text{confining} & \mathbb{Z}_N^{(1)} \text{ unbroken} \\ \langle W(C) \rangle \sim e^{-\mu \cdot \text{Perimeter}(C)} & \text{deconfined} & \mathbb{Z}_N^{(1)} \text{ broken} \end{array}$$

Same class:  $\mathbb{Z}_N$  gauge theory (Example 2),  $U(1)$  with only charge- $N$  matter (endpoint criterion).  
Perfect segue into Part 6: generalized order parameters.

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# Generalized order parameters

|                | 0-form                                                                           | 1-form                                                   |
|----------------|----------------------------------------------------------------------------------|----------------------------------------------------------|
| charged object | local $\mathcal{O}(x)$                                                           | loop $W(C)$                                              |
| unbroken       | $\langle \mathcal{O}^\dagger(x) \mathcal{O}(0) \rangle \sim e^{-m x }$           | $\langle W(C) \rangle \sim e^{-T \text{Area}(C)}$        |
| broken         | $\langle \mathcal{O}^\dagger(x) \mathcal{O}(0) \rangle \rightarrow \text{const}$ | $\langle W(C) \rangle \sim e^{-\mu \text{Perimeter}(C)}$ |

- Perimeter term = local line counterterm; after removing it, a perimeter law is the higher-form analogue of long-range order.
- General theorem: charged expectation value  $\Leftrightarrow$  state not symmetric — valid for any form degree.

# The photon is a Goldstone boson

Goldstone logic, one form-degree up: couple the current to its background and demand invariance under  $a \rightarrow a + \lambda_1$ ,  $B \rightarrow B - \mathrm{d}\lambda_1$ :

$$\mathcal{L}_{\text{eff}} = -\frac{1}{2g^2}(\mathrm{d}a + B) \wedge \star(\mathrm{d}a + B) \quad \xrightarrow{B=0} \quad S = -\frac{1}{2g^2} \int \mathrm{d}a \wedge \star \mathrm{d}a$$

— free Maxwell theory.

## Punchline

The photon's masslessness is protected by a spontaneously broken 1-form symmetry — a *symmetry* statement, not gauge redundancy.

# Higher-form Coleman–Mermin–Wagner

Goldstone fluctuations of a flat charged  $p$ -brane of size  $L$ : only the  $n = D - p$  *transverse* momenta contribute,

$$\langle W(C) \rangle \sim \exp \left[ - \frac{g^2}{2} L^p \int_{1/L}^{\Lambda} \frac{d^n k_{\perp}}{(2\pi)^n} \frac{1}{k_{\perp}^2} \right] \quad \Rightarrow \quad \boxed{\text{no continuous } p\text{-form SSB for } D - p \leq 2.}$$

- $p = 0$ ,  $D = 2$ : ordinary Mermin–Wagner / Coleman.
- $p = 1$ ,  $D = 3$ : marginal — monopole instantons kill the SSB (Polyakov confinement).
- $p = 1$ ,  $D = 4$ : allowed — the Coulomb phase and its photon.

## Topological order = higher-form SSB... almost

Local indistinguishability  $\langle \text{gs}_a | O | \text{gs}_b \rangle = c_O \delta_{ab}$  splits into *diagonal* and *off-diagonal* conditions. If  $|\text{gs}_1\rangle = W_C |\text{gs}_2\rangle$  with  $W_C$  topological, deforming  $W_C$  off the support of  $O$  gives the diagonal condition. **1-form SSB alone does not give the off-diagonal one.**

### What is missing: a second topological operator

A topological  $V_{\hat{C}}$  with distinct eigenvalues on the two states forces  $\langle \text{gs}_2 | O | \text{gs}_1 \rangle = 0. \Rightarrow$   
**topological order = 1-form SSB + a mutually anomalous partner symmetry.**

Counterexample (McGreevy v3): deformed toric code — beyond  $\beta_c = \frac{1}{2} \ln(1 + \sqrt{2})$  the topological entanglement entropy vanishes, yet the electric loop keeps a perimeter law.

# 't Hooft anomalies of higher-form symmetries

**Anomaly** = obstruction to gauging = RG-invariant phase of  $Z[B]$ ; representable by inflow, e.g.

$$e^{iS_{\text{inflow}}} = \exp\left(\frac{2\pi i}{N} \int_{Y_{D+1}} B_{p+1} \smile C_{D-p}\right).$$

- **Maxwell:**  $U(1)^e \times U(1)^m$  mixed anomaly — gauge either, never both.
- **Anyon statistics = anomaly data:** gauging a 1-form symmetry in  $D = 3$  = condensing the anyon; nontrivial spin or braiding obstructs it — the “partner operator” of the previous slide.
- $SU(N)$  at  $\theta = \pi$ : mixed  $\mathbb{Z}_N^{(1)}$ -CP anomaly  $\Rightarrow$  no trivially gapped symmetric vacuum.

# The enlarged Landau paradigm

| phenomenon                | generalized-symmetry description      |
|---------------------------|---------------------------------------|
| deconfinement             | SSB of electric 1-form symmetry       |
| photon / Coulomb phase    | Goldstone of broken 1-form $U(1)$     |
| abelian topological order | 1-form SSB + mixed anomaly (braiding) |
| SPT edge modes            | boundary must match bulk anomaly      |
| anyon statistics          | higher-form 't Hooft anomaly data     |

## McGreevy's proposal

Phases are characterized by *generalized* symmetries and their anomalies; “beyond Landau” largely means “Landau, with extended operators.”

- 1 Motivation
- 2 Mathematical foundation
- 3 Ordinary symmetry, recast
- 4 Higher-form symmetry: definition
- 5 Examples
- 6 Spontaneous breaking and phenomena
- 7 Outlook**

# Beyond invertible higher-form symmetry

- **Non-invertible / categorical**: drop the group law — e.g. the Kramers–Wannier defect of critical Ising,  $\mathcal{N}^2 = 1 + \eta$ : no inverse. Symmetries form a fusion category.
- **Subsystem symmetries**: rigid (non-deformable) supports  $\Rightarrow$  fractons.
- **Higher groups**: 0-form and 1-form parts mix.
- **SymTFT**: all symmetry + anomaly data as a  $(D+1)$ -dimensional topological bulk.

## Summary

- ① A symmetry is a **topological operator**; conservation = topological invariance.
- ②  **$p$ -form symmetry**: codimension- $(p+1)$  operators,  $p$ -dimensional charges, measured by **linking**; abelian for  $p \geq 1$ .
- ③ Maxwell:  $U(1)^e \times U(1)^m$ ; toric code /  $\mathbb{Z}_N$  gauge theory realize the linking algebra exactly;  
Chern–Simons: the anyon lines generate their own (anomalous) 1-form symmetry;  
Yang–Mills:  $\mathbb{Z}_N^{(1)}$  center.
- ④ Area vs. perimeter law; photon = Goldstone; no continuous SSB for  $D - p \leq 2$ ;  
topological order = SSB + anomaly.

## References

GKSW, arXiv:1412.5148 — McGreevy, arXiv:2204.03045 — Iqbal, arXiv:2407.20815 — Córdova–Dumitrescu–Intriligator–Shao, arXiv:2205.09545.

*Thank You!*